Jeet Sharma

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Summary

Software engineer building production backend systems and data intensive platforms, with experience in distributed services, large scale storage design, and cloud deployed infrastructure. Developed scalable ingestion, indexing, and analysis systems supporting real world investigative workloads from architecture to production

Education

University of Massachusetts, Amherst, MA, United States — Sep 2024 – May 2026
Master of Science in Computer Science — GPA: 3.86/4.00
University of Mumbai, Mumbai, India — Aug 2019 – May 2023
Bachelor of Engineering in Computer Engineering — GPA: 3.87/4.00

Professional Experience

Software Engineering Intern | Karpuragaurai Technologies, India — May 2025 – Aug 2025
Tech Stack: Python, Redis, REST APIs, AWS, Docker, Celery

- Implemented semantic retrieval using PostgreSQL and PGVector to index client knowledge bases for grounded LLM responses
- Deployed containerized backend services on AWS ECS behind an Application Load Balancer with autoscaling workers
- Added Redis-based response caching and asynchronous background processing to reduce end to end query latency by 30%

Software Engineer | Lab Systems (I) Pvt. Ltd., India — Aug 2022 – Aug 2024
Tech Stack: NextJS, Python, C++, Flask, SQL/NoSQL, Cloudflare, Docker, Software Architecture, Solution Designing, AWS, Node.js, REST APIs, MongoDB, LevelDB, Linux, Web Security, Distributed Systems, System Design

- Designed and implemented the distributed backend architecture for a digital forensics platform, enabling scalable ingestion and processing of multi terabyte investigative datasets in production
- Built high performance backend services in Python and C++ for data ingestion, indexing, and search, supporting low latency queries and concurrent investigative workloads
- Engineered a hybrid MongoDB and LevelDB storage architecture enabling indexed queries across datasets exceeding 2B blockchain transaction records
- Developed distributed analysis pipelines and a production cryptocurrency investigation system, reducing manual investigation time by 70% and improving case processing throughput
- Designed REST APIs and asynchronous job processing to support reliable execution of compute intensive workflows at scale
- Improved production reliability and system correctness by implementing automated testing, validation pipelines, and containerized Docker based cloud deployments on Linux infrastructure

Skills

Languages: Python, C++, JavaScript, TypeScript
Backend & Data: REST APIs, PostgreSQL, MongoDB, Redis, Elasticsearch, PGVector
Frontend: React, Next.js, Tailwind CSS
Cloud & Infrastructure: AWS (ECS, EC2, ALB), Docker, Linux, CI/CD
AI Systems: LLM integration, RAG pipelines, embeddings, vector search
Testing & Practices: Unit testing (PyTest, Mocha), system design fundamentals

Relevant Projects

LLM-Router: Intelligent LLM Selection and Cost Optimization | Tech: PyTorch, DistilBERT, CodeBERT — Project Link

- Fine tuned DistilBERT and CodeBERT classifiers on real world prompt response data to dynamically route requests
- Reduced inference costs by 70% by reducing reliance on expensive models like GPT-4o while maintaining high output quality

Patent

D Ein Blockchain-basiertes Medizinlogistiksystem (Granted)
German Patent File No - 20 2023 102 823.3 — Date: 19 Sep 2023
Developed a web application for tracking prescription drugs using Blockchain, enabling secure and transparent pharmaceutical supply chain management with improved traceability by 80% and significantly reduced instances of counterfeit drugs in deployments